

Correlation Illusion Cards Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	correlation-illusion-cards

Big idea: Patterns can feel meaningful before they are mathematically justified.

Overview

Students sort paired-variable cards and discover how small samples and salient cases create illusory correlations.

Student Prompt

Inspect Correlation Illusion Cards Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Correlation Illusion Cards Stimulus A	Student-facing stimulus 1: Cards pair 'used flashcards / did not use flashcards' with 'high quiz score / lower quiz score'. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Correlation Illusion Cards Stimulus B	Student-facing stimulus 2: A few vivid cards show flashcards plus high scores, while the full set is mixed. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Correlation Illusion Cards Reveal / Twist	Student-facing stimulus 3: A few vivid cards show flashcards plus high scores, while the full set is mixed. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Patterns can feel meaningful before they are mathematically justified.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Correlation Illusion Cards Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Correlation Illusion Cards Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	When does a perceived pattern become evidence?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how patterns can feel meaningful before they are mathematically justified.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Build card sets where confirming cases are memorable but not actually dominant.
- Ask for a gut judgement before students count the full table.

Ready-to-Use Examples

- Cards pair 'used flashcards / did not use flashcards' with 'high quiz score / lower quiz score'.
- A few vivid cards show flashcards plus high scores, while the full set is mixed.

Suggested Timing

Stage	Teacher move	Watch for
1	Show a small set of memorable paired examples and ask students to guess the relationship.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Give groups the full card set to sort into a four-cell table.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Students count all cells, not just confirming cases.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups decide whether the perceived relationship survives counting.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief why humans notice some pairings more than others.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- A memorable case is not automatically representative evidence.
- Which cell of the table did your intuition ignore?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Support: color-code the four table cells.
- Challenge: ask students to estimate strength of association before and after counting.

Extension Tasks

- Students collect a harmless class dataset and test for perceived versus counted patterns.
- Connect to stereotypes, health claims, or study advice.

Related Activities

- Proof versus Pattern
- P-hacking with Random Data
- Reliability Ranking